

CUSTOMER CHURN PREDICTION

Internship Project Report

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Organization

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Internship Project

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I would like to express my sincere gratitude to PERSEVEX for providing me with the opportunity to work on the Customer Churn Prediction internship project. This project provided valuable practical exposure to data analysis, data visualization, Python programming, and the development of an interactive dashboard.

I am thankful for the learning experience gained during the project and for the opportunity to apply technical concepts to a customer analytics problem.

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1. INTRODUCTION

Customer churn refers to customers discontinuing or leaving a service. Understanding churn is important for businesses because identifying churn-related patterns can support customer retention and better business decisions. The Customer Churn Prediction project focuses on analyzing customer information and presenting churn-related patterns through an interactive dashboard.

The project uses Python and Streamlit to load customer data, calculate basic dataset statistics, display customer records, analyze the churn distribution, and examine the distribution of numerical features such as tenure in months.

2. PROJECT OBJECTIVES

- To analyze customer data in a structured and understandable manner.
- To identify and display the overall distribution of customer churn.
- To calculate basic churn statistics such as the number of churned customers and churn rate.
- To provide an interactive view of customer records.
- To analyze numerical customer features using visualizations.
- To develop a simple and user-friendly dashboard using Streamlit.

3. PROJECT OVERVIEW

The completed application is an interactive Customer Churn Prediction Dashboard. The dashboard contains four main sections: Overview, Customer Data, Churn Analysis, and Feature Analysis. A sidebar menu is provided for navigation between these sections.

Dashboard Metric	Value
Total Customers	1000
Total Features	25
Churned Customers	78
Churn Rate	7.80%

4. TECHNOLOGIES AND TOOLS USED

Technology / Tool	Purpose
Python	Programming language used to build the application.
Pandas	Used for loading and working with tabular customer data.
Streamlit	Used to create the interactive web dashboard.
Operating System / File Handling	Used to construct the dataset path and check whether the CSV file exists.
CSV Dataset	Customer data source used by the dashboard.

5. DATASET DESCRIPTION

The dashboard displays a customer dataset containing 1,000 customer records and 25 features. The visible dataset preview includes fields such as `customer_id`, `industry`, `company_size`, `geography`, `subscription_tier`, `tenure_months`, `MRR`, and `billing_frequency`. The application reads the dataset from a CSV file.

5.1 Example Dataset Fields

Field	Description
<code>customer_id</code>	Unique customer identifier.
<code>industry</code>	Customer industry category.
<code>company_size</code>	Size category of the customer company.
<code>geography</code>	Geographical location/category.
<code>subscription_tier</code>	Subscription plan or tier.
<code>tenure_months</code>	Number of months the customer has remained with the service.
<code>MRR</code>	Monthly recurring revenue value.
<code>billing_frequency</code>	Billing frequency such as monthly or annual.

6. DATA LOADING AND VALIDATION

The application constructs the dataset path using the application's directory and the data folder. Before reading the CSV file, it checks whether the expected dataset exists. If the file is not found, the application displays an error message and stops. The CSV file is then loaded into a Pandas DataFrame.

The dashboard also calculates dataset-level information such as the total number of records, total number of columns, missing values, and duplicate rows when the churn column is not detected.

7. EXPLORATORY DATA ANALYSIS

Exploratory analysis is presented through summary metrics and visualizations. The application previews the first 10 records and provides dataset information. For numerical features, the Feature Analysis section allows a numerical column to be selected and displays its distribution.

8. CUSTOMER CHURN DASHBOARD

The dashboard provides a simple navigation system through a sidebar. Users can select Overview, Customer Data, Churn Analysis, or Feature Analysis. This structure makes the analysis easier to access without requiring users to work directly with code.

9. DASHBOARD MODULES

9.1 Overview

The Overview page displays the Dataset Overview, total customers, total features, churned customers, churn rate, and a dataset preview. In the completed dashboard, the displayed values are 1,000 customers, 25 features, 78 churned customers, and a 7.80% churn rate.

9.2 Customer Data

The Customer Data section displays customer records in a tabular format. It provides a direct view of the customer information available in the dataset.

9.3 Churn Analysis

The Churn Analysis section searches the dataset for a standard churn-related column. It recognizes names such as churn, churned, is_churn, and customer_churn. When a churn column is found, its value counts are displayed as a bar chart.

9.4 Feature Analysis

The Feature Analysis section allows the user to select a numerical feature. The selected feature is visualized as a distribution chart. The completed dashboard shows the distribution of tenure_months as an example.

10. CHURN ANALYSIS

The Churn Analysis module identifies the churn column dynamically instead of depending on one fixed column name. The values are converted to lowercase text and the application counts recognized churn values such as yes, 1, true, churn, and churned. The churn rate is calculated as the churn count divided by the total number of customers and multiplied by 100.

$$\text{Churn Rate} = (\text{Number of Churned Customers} / \text{Total Customers}) \times 100$$

For the displayed dashboard, 78 churned customers are reported out of 1,000 customers, resulting in a displayed churn rate of 7.80%.

11. FEATURE ANALYSIS

Feature Analysis provides a numerical feature selector and visual distribution. The completed dashboard demonstrates this functionality using tenure_months. The chart allows the user to visually inspect how customer tenure values are distributed.

12. RESULTS AND OBSERVATIONS

- The dashboard successfully loads and presents a customer dataset containing 1,000 records.

- The dataset contains 25 features.
- The dashboard reports 78 churned customers.
- The displayed churn rate is 7.80%.
- Customer records can be viewed through the Customer Data section.
- Churn distribution is visualized using a bar chart.
- Numerical feature distributions can be explored interactively.
- The sidebar provides simple navigation among the four dashboard sections.

13. ADVANTAGES OF THE PROJECT

- Easy-to-use interactive dashboard.
- Clear presentation of important customer metrics.
- Interactive exploration of customer data.
- Visual representation of churn distribution.
- Numerical feature analysis through selectable visualizations.
- Basic dataset validation and error handling.
- Can help users understand customer churn patterns more quickly than raw tables alone.

14. LIMITATIONS

- The dashboard analysis depends on the quality and structure of the CSV dataset.
- The visible dashboard focuses on descriptive analysis and visualization.
- A churn prediction model and its evaluation metrics are not documented in the supplied dashboard screenshots/code excerpts.
- The displayed churn rate describes the current dataset and should not automatically be interpreted as a future churn forecast.

15. FUTURE SCOPE

- Integrate a trained machine-learning classification model for individual customer churn prediction.
- Add model evaluation metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix when a prediction model is implemented.
- Add filters for industry, geography, subscription tier, company size, and billing frequency.
- Add customer-level risk categories such as low, medium, and high churn risk.
- Add downloadable reports and filtered datasets.
- Add trend analysis and additional interactive charts.

16. CONCLUSION

The Customer Churn Prediction project demonstrates how customer data can be organized, analyzed, and presented through an interactive Streamlit dashboard. The completed application provides four main modules—Overview, Customer Data, Churn Analysis, and Feature Analysis—and presents useful summary information about the dataset. With 1,000 customers, 25 features, 78 displayed churned customers, and a 7.80% churn rate, the dashboard provides a clear starting point for understanding churn-related patterns. The project can be further extended by integrating a machine-learning prediction model and more advanced customer-risk analytics.

PROJECT SUMMARY

Item	Details
Project	Customer Churn Prediction
Student	Charu Gupta
Organization	Persevex
Project Type	Internship Project
Dataset Size	1,000 customers
Number of Features	25
Displayed Churned Customers	78
Displayed Churn Rate	7.80%
Dashboard	Streamlit-based interactive dashboard